

SPSS 4 Correlation Analysis

Topics

- Correlation
 - Relationship of 2 variables
 - Prediction accuracy
- Pearson r
- Correlation matrix
 - Relationship among multiple variables
- Correlation assumptions
- Partial Correlation

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Hypothesis Of Association: Correlation

- Correlation of variable A and B
 - Measure how often A and B change together linearly
- Correlation is **NOT** cause and effect:
 - Difficult to predict the direction of the relationship
 - A → B
 - B → A
 - X → A + B
 - Difficult to isolate the real cause
 - Only used to make better than chance prediction

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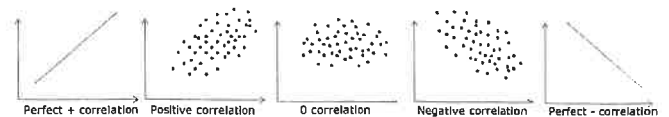
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Pearson correlation > 0.999 ?
correlation ~ 1

Correlation in scatter plot

- **Positive correlation**
 - oval shape from lower-left to upper-right
- **Negative correlation**
 - oval shape from upper-left to lower-right
- **Perfect positive/negative correlation**
 - a straight line
- **0 correlation**
 - no direction
- Correlation measure how tightly the data cluster around the regression line

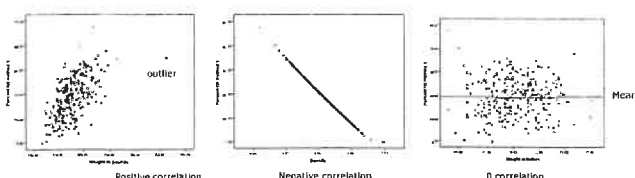
Notes: 2 variables may have 0 correlation but have non-linear relationship (U or S shaped curve)



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Task: Scatterplot

- Scatterplot of body fat percentage and weight for all samples
 1. Double click file "bodyfat.sav" to open
 2. Graphs > Legacy Dialogs > Scatter/Dot > Simple Scatter > Define
 3. Set Y Axis to fatpct1, set X Axis to weight > OK
- Try scatter plot for body fat percentage and body density
- Try scatter plot for body fat percentage and height
- To draw mean reference line:
 - Double Click The Plot To Edit > Options > Y Axis Reference Line > Set to choose Mean > Apply > Close



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Pearson r

- A number that indicates the strength of **linear relationship** between 2 sets of samples in **interval form** and with **normal distribution**
- Measure how much they **both vary** from average
- Calculating r:
 - Mean of the products of standard scores of all samples

$$r = \frac{\sum z_x z_y}{N - 1}$$

- r ranges from -1 (perfect -ve) to 1 (perfect +ve). 0 for no correlation

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Task: Paired t test for 2 measures of body fat

- Paired t test for 2 method to measure body fat
 - Analyze > Compare Means > Paired Samples T Test
 - Set Paired Variables List to fatpct1, fatpct2 > OK

Interpretation of the results:

- Correlation

Paired Samples Statistics					
Pair	Percent fat method 1	Mean	N	Std. Deviation	Std. Error Mean
1	Percent fat method 1	19.1565	252	6.38077	.52794
2	Percent fat method 2	18.9255	252	7.74922	.48815

Paired Samples Correlations			
Pair	Percent fat method 1 & Percent fat method 2	N	Correlation
1	Percent fat method 1 & Percent fat method 2	252	.908

The 2 methods have significantly high positive correlation.

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Task: Paired t test for 2 measures of body fat (Con't)

- Difference between the two samples
- Note: The two methods have high correlation, but the means of them are different. It indicates that method 1 results in higher fat value, and the samples having higher fat reading with method 1 tend to be higher in method 2.

Large t indicates large difference between the means

Paired Samples Test									
Paired Differences									
Pair	Percent fat method 1 - Percent fat method 2	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		t	df	Sig. (2-tailed)
1	Percent fat method 1 - Percent fat method 2	.22184	.83677	.04013	Lower	Upper	5.511	251	.000

The mean of Method 1 is significantly greater than Method 2

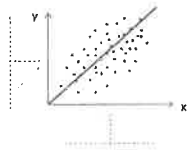
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Sig. level $\Rightarrow p = .000$
 < 0.05 statistically different

Requirements for using Pearson r

(check assumption)

- Requirements for using Pearson r
 - Interval or ratio data (scale data)
 - ✓ Random sample selection (independent sample) *collector can check*
 - Each variable distribution must approach normality $\Rightarrow$ Histogram
 - Variances along the best-fit line are similar (homoscedasticity)
 - The association between 2 variables must be linear, not curvilinear relationship. Check scatter plot

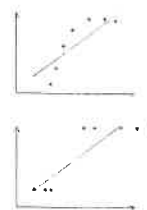
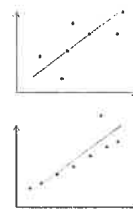


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Weakness of correlation

- Many different relationships of the 2 variables can result in the same correlation coefficient
- 2 significantly correlated variables may have no real relationship at all
- Always plot the data to confirm that correlation summarizes the relationship between 2 variables well



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Partial Correlation Coefficients

- Measure the correlation between variables while the effects of other variables are kept constant
- eg. Measure the correlation between percentage of fat and weight, while keeping the effect of height constant (controlling for height)
- Interpretation of sig. level is the same as correlation coefficient

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Partial Correlation Coefficients

- Partial correlation coefficients and observed significance
 - Analyze > Correlate > Partial
 - Set Variables to fatpct1, weight
 - > Options > Zero-order Correlations > Continue
 - Set Controlling For to height

The first half of the output is the ordinary correlation matrix

Correlations				
Control Variables				
Percent fat method 1	Weight in pounds	Height in inches		
Correlation	.613	-.025		
Significance (2-tailed)	.000	.894		
df	250	250		
Weight in pounds	Height in inches			
Correlation	.613	.000		
Significance (2-tailed)	.000	.900		
df	250	250		
Height in inches				
Correlation	-.025	.487		
Significance (2-tailed)	.894	.000		
df	250	250		

Recall that height is significantly correlated to weight, but not to percentage of fat

weak corr.

Strong corr.

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